

In situ microscopy of composite Germanium/Germanium oxide microneedle growth

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ABSTRACT

Composite germanium/germanium oxide needles with a carcass/filler structure, thicknesses of 0.25–10 μm and lengths up to 450 μm were grown by chemical transport involving gaseous GeO . The synthesis was carried out at temperatures of 897–1100 K in a newly developed miniature cell. Through a top window one can observe the growth process in situ via video microscopy. The cell can be evacuated and sealed, enclosing all products generated in reactions, but can easily be opened afterwards for an ex situ analysis of the needles. The in situ observations show that the needles grow with longitudinal velocities of up to 30 $\mu\text{m/s}$, depending on temperature and needle thickness, and with lateral velocities of up to 0.11 $\mu\text{m/s}$. The surface structure and the cross-sections of the needles were studied ex situ by SEM as well as EDX spectroscopy combined with element mapping. The surface structure of needles exhibits highly regular arrays of germanium/germanium oxide microunits but also shows irregularities and defects. The study of the longitudinal section and the lamella showed that this regularity is remarkable also inside the needle all way down to the core. Furthermore, according to the SAED and Raman spectroscopy, the needle carcass (core with branches) is mainly composed of a cubic Ge and an amorphous filling mass of Ge and O, which presumably corresponds to GeO_2 .

1. Introduction

Until the mid-1960 s, germanium (Ge) was the dominant elemental semiconductor material. Ge was preferred by most design engineers for computer logic circuits until it was replaced by silicon (Si) as the basic material for transistors. Due to their properties, Si and its native insulating oxide (SiO_2) were the most suitable components for electronic switching devices made with technology available during that time [1]. In recent decades, due to advantages in technology allowing the miniaturisation of electronic components, the trend is changing and Ge is once again beginning to attract research interest. For example, the mobilities of electrons and holes in Ge are significantly higher than those in Si, which makes it more advantageous for the use in metal-oxide semiconductor field-effect transistors (MOSFETs) [2]. However, the native Ge oxide (GeO_2) does not have sufficient chemical and thermal stability, which manifests itself, e.g., in higher defect densities of Ge/

GeO_2 interface than of Si/ SiO_2 [3]. While this instability is a disadvantage for Ge-based microelectronics, in the case of producing Ge quantum dots in insulating GeO_2 matrix [4,5], for example, it is not only desirable, but a necessity. Therefore, studying the behaviour of a Ge- GeO_2 system, in which both components are in the same or different physical state, under a variety of conditions, is a source not only for new discoveries, but also for the potential of their applications.

Some of the properties of the Ge- GeO_2 system were known through research in the early second half of the 20th century [6–11]. In the early studies of thermodynamic aspects of this system, for example, the germanium-oxygen (Ge-O) phase diagram was determined experimentally [6,7]. According to it, Ge and GeO_2 can form an alloy. Trumbore et al. [6] showed the co-existence of two phase diagrams each with an eutectic point: the stable phase diagram for Ge and tetragonal GeO_2 , in which the eutectic is characterised by a temperature of approx. 1185 K and a GeO_2 mole fraction of $x_{\text{GeO}_2(\text{tet})} = 0.87$, and the metastable phase

Abbreviations: BN, boron nitride; CMOS, complementary metal-oxide-semiconductors; EDX, energy-dispersive X-ray; FE-SEM, field emission scanning electron microscopy; FIB, focused ion beam; FS, fused silica; MOSFET, metal-oxide semiconductor field-effect transistor; SAED, selected area electron diffraction; STEM, scanning transmission electron microscopy; TC, thermocouple; VLS, vapor-liquid-solid; VSS, vapor-solid-solid.

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diagram for Ge and hexagonal GeO₂ with the eutectic at about 1143 K and $x_{\text{GeO}_2(\text{hex})} = 0.9$. In both aforementioned articles, there is no information on the composition or purity of materials – GeO₂ and Ge – used. Therefore, it is not possible to make an unequivocal assessment on the accuracy of the results obtained. Moreover, there are very few articles that dealt with the experimental construction of the Ge–O phase diagram, which makes it difficult to evaluate the findings, so Dinsdale et al. [12] derived a calculated phase diagram based on known experimental data. According to it, the stable eutectic is characterised by approx. 1189 K and $x_{\text{GeO}_2(\text{tet})} = 0.71$, whereas the eutectic of the metastable phase diagram corresponds to 1127 K and $x_{\text{GeO}_2(\text{hex})} = 0.765$.

Due to other early intensive investigations on the Ge–GeO₂ system [9–11], it was found that annealing of a mixture of solid Ge and GeO₂ results in the formation of volatile germanium monoxide (GeO) according to the reaction



Direct observation of Ge/GeO₂ stacks showed Ge consumption during GeO desorption, and confirmed that the latter is generated due to reaction (1) [13]. Complementary metal-oxide-semiconductors (CMOS) are used in most today's computer microchips [14] and, therefore, understanding and controlling the GeO desorption from a GeO₂/Ge stack are of vital importance for Ge MOSFETs [15]. Prabhakaran et al. [16] showed, that on annealing ultrathin GeO₂ layer on Ge, GeO₂ species transform to GeO, which remain on the surface up to approx. 693 K and desorb as GeO(g) at higher temperatures. Wang et al. [15] suggested from observations on isotope tracing experiments, that O of desorbed GeO is not from the Ge/GeO₂ interface, but from the top layer consisting of GeO₂, and propose an O vacancy model of the GeO desorption mechanism.

For other applications, volatile GeO is the source for production of thin homogeneous GeO(s) films by deposition of vapor molecules from supersaturated GeO(g) on a substrate at temperatures of 673–823 K or even at room temperature according to the reaction [4,5]



GeO is an attractive species for the production of memristors [17], quantum dots [5] and GeO₂ nanowires [18], because of its metastability in the solid state. The GeO(s) film readily decomposes into Ge and GeO₂ in several minutes at temperatures above 573 K in accordance with the following reaction equation [4]



The disproportionation (3) was observed experimentally by X-ray photoelectron spectroscopy and this result was also confirmed by Raman spectroscopy [19]. Through the reaction (3), the system originally consisting of mixture of solid Ge and GeO₂, which was the basis for GeO(g) production by the reaction (1), returns to its original state, but with different properties. Here, GeO₂ forms a glassy matrix, with excess Ge atoms being segregated as Ge nanoclusters [4]. Xie et al. [20] showed that an amorphous GeO₂ can be locally crystallised into a hexagonal structure on Ge due to interface reaction at the GeO₂/Ge interface at 933 K. It was reported in [21], that Ge/GeO₂ microcrystals with cubic Ge core and hexagonal GeO₂ shell structure can be considered a promising candidate for photon-free degradation of dyes.

Heating the Ge/GeO₂ heterostructure at a temperature of 823–873 K for several minutes results in its complete evaporation in accordance with the reaction (1), which was proved by scanning ellipsometry [22]. Therefore, it can be concluded that through the reactions (1)–(3) not only volatile GeO, metastable GeO(s) film or heterostructure, consisting of GeO₂ matrix and segregated Ge, can be produced depending on conditions, but also in a closed system it is possible to turn reactions (1)–(3) into a cycle which can be controlled only by heating or cooling the system.

In summary, the disproportionation (3) can hence be used to

generate randomly dispersed Ge nanoclusters in a GeO₂ matrix. An interesting question is, however, if one could also use it to crystallise ordered Ge nanocrystal arrays in GeO₂. We have, therefore, investigated the reaction (3) under well-controlled thermochemical conditions in a closed reaction cell with a volume of a few cubic millimeters, containing up to a several hundred micrograms of Ge and hexagonal GeO₂. Instead of a thin film of Ge nanocrystal arrays we discovered the formation of composite sub-micrometer- and micrometer-sized germanium/germanium oxide needles formed by chemical transport. This newly type of germanium/germanium oxide microstructures can exhibit highly regular arrays of microunits but also shows irregularities and defects. We succeeded in observing the growth of such needles in the reactor cell in situ by video microscopy and studying them ex situ by use of field emission scanning electron microscopy (FE-SEM), energy-dispersive X-ray (EDX) spectroscopy combined with element mapping, scanning transmission electron microscopy (STEM), selected area electron diffraction (SAED), and Raman spectroscopy.

2. Experimental setup

2.1. Apparatus

The central part of the micro reactor is similar to one already described in our previous studies, e.g., in [23]. In this study illumination of a chemical transport cell is provided not only from above but also from below, as shown in Fig. 1 a. The chemical transport cell is composed of three glass plates which are stacked on top of each other and has planar dimensions of 10 mm × 10 mm and a thickness of 1.7 mm. The middle one acts as a spacer and is made of Corning® EAGLE XG® (alkaline earth boro-aluminosilicate) glass with a softening point (10^{7.6} poise) of 1244 K [24]. A hole of a diameter of approx. 3.5 mm is drilled in the centre of the spacer. Two fused silica (FS) plates, which have a much higher softening point than that of the Corning® glass, cover the spacer from above and below. This provides a small reaction chamber with a volume of approx. 7 mm³, which can be filled with reagents – Ge and GeO₂. The cell becomes sealed once it is heated to a temperature of 973–1073 K. It should be noted, that Corning® glass contains in addition to the basic glass compound SiO₂, also major elemental constituents aluminium (Al), boron (B), calcium (Ca), magnesium (Mg) and strontium (Sr) as well as some trace components [29], which can be phase separated from the glass at high temperatures and react with the reagents or catalyse their reaction under certain conditions. But the glass used is so far the only one that is suitable to keep the cell sealed in the temperature range of 1138–1253 K required in our experiments. In the case of GeO(g) or Ge(g) involved in chemical transport according to reaction (1) and (4) respectively, it is easy to recognise whether the cell has been sealed or not. In the latter case they leave the cell and are observed as orange-brown films on the observation window in the vacuum chamber below the microscope objective shown in Fig. 1 a. Therefore, the presence or absence of this layer, which is confirmed with EDX spectroscopy to be composed of Ge and O, serves as an indicator of sealing condition. If the cell remained unsealed, no needle growth was observed in it.

The glass cell is sandwiched between two boron nitride (BN) plates, which have dimensions of approx. 10 mm × 10 mm × 1 mm and a small drilled hole in the centre of one narrow side, so that a thermocouple (TC) of type C can be inserted in each plate. They hold both TC in their proper position and are designated as BN TC plates in the following text. The BN frame, the BN cover and the BN bottom fix the cell and the BN TC plates. The cell can be heated by use of two BORAELECTRIC® heaters (Momentive Performance Materials Inc.). Each of them is driven by a separate DC power supply unit (FUG NTN 1400–65). By controlling the applied voltage for each heater, it is possible to provide a stable and accurate temperature difference between the FS plates in the cell. The whole arrangement is operated in a high vacuum chamber at a pressure of 10^{−7}–10^{−6} mbar, which is reached by means of a turbomolecular pump

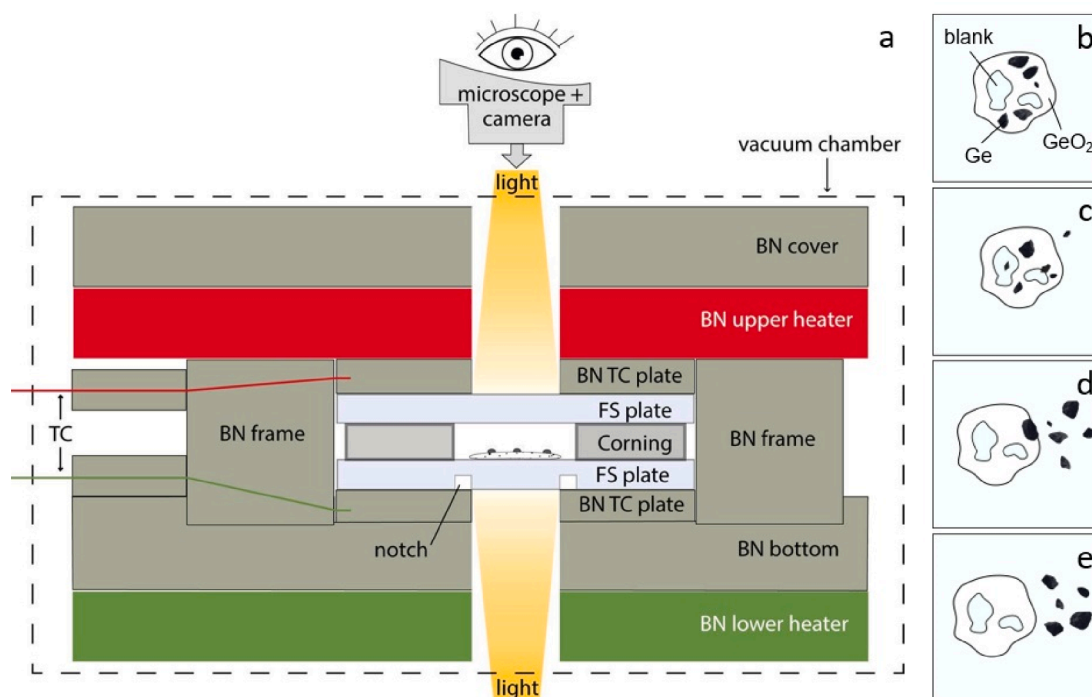


Fig. 1. **a** Vertical cross-section of the central part of the micro reactor for chemical transport reactions. All details regarding the device components and cell preparation are given in the text. The cell shown here has a contact arrangement of the reagents – tiny Ge splinters on a small pile of hexagonal GeO_2 powder. BN stands for boron nitride, FS for fused silica and TC for thermocouple. **b–e** Top view of various reagent arrangements in the centre of the bottom FS plate that have been used in experiments: **b** full contact case (all splinters on powder), **c** partially contact case (splinters on powder, on blank regions within pile, which are uncovered by powder, or on bare FS plate surface), **d** quasi-non-contact case (only one splinter in contact with powder), and **e** non-contact case (reagents lie separately from each other). The GeO_2 powder layer has a width of about 2 mm. Its thickness can vary and some regions of the pile can be uncovered by the powder. Most of the experiments in the present study were carried out in a partially contact case (**c**).

(Edwards R72202003) backed by a scroll pump (Varian SH-100). The BN cover and bottom, the heaters, and the BN TC plates have a central opening of approx. 2 mm for illumination of the cell from above and below. Moreover, through the upper opening the processes in the cell can be observed in situ via a video microscope (microscope Olympus SZX16 with camera AVT Mako G-192C). The LabVIEW (National Instruments) software is used to control the power supplies and to measure the temperature of the FS plates as a function of time by means of TC.

2.2. Preparation of reaction cells

The 0.7 mm thick polished Corning® EAGLE XG® plate (MTI Corporation) has initial size of 10 mm × 10 mm. Then its planar dimensions are reduced to approx. 9 mm × 9 mm using a diamond tool. This is done to ensure that at high temperatures, when it softens and spreads slightly apart, it does not come into contact with the BN frame, which makes it easier to remove the cell after solidification of the spacer. Afterwards, a hole with the diameter of approx. 3.5 mm is drilled in the middle of it by use of a diamond tool. Two 0.5 mm thick polished FS plates (MTI Corporation) with a size of 10 mm × 10 mm are used to cover the spacer from above and below to create a small reaction chamber. A circular notch with an average diameter of 3.5 mm, a width of 0.6 mm and a depth of 0.25 mm is grinded around the centre of one of the fused silica plates. This creates a small platform-like area. The notch makes it easier to open the cell after the experiment. The second FS plate remains unstructured. All three components of the cell are cleaned in acetone using an ultrasonic bath for around 20 min, flushed with double distilled water, dried under a strong blast of air and stored in a drying cabinet at 200 °C for at least 10 h to remove adsorbed water and other gases.

The reagents used in our study are hexagonal GeO_2 powder ($\geq 99.99\%$, Sigma-Aldrich) and Ge splinters. The latter are produced by breaking monocrystalline Ge substrates ($>99.9953\%$, undoped, MTI

Corporation) into small pieces. A mass ratio of Ge to GeO_2 of approx. 1:2 is used. First, four to eight Ge splinters, each with a mass of 25–70 μg , are weighted for a total mass of about 200 μg . Also, approx. 400 μg of hexagonal GeO_2 powder, which is stored permanently in a drying cabinet at 200 °C, are placed in the centre of the lower FS plate and distributed with a spatula in a layer with a width of about 2 mm. The thickness of the layer can vary, and the layer is also nonuniform, i.e., some regions can be uncovered by the powder. Ge splinters are then placed either on the powder or near it. Fig. 1 **b–e** shows the possible arrangements of the reagents: full contact case (**b**), where all splinters are on the powder, non-contact case (**e**), where Ge and GeO_2 are separately arranged, and some intermediate cases (**c** and **d**), in which some splinters are either on or near the powder. The vast majority of the experiments in the present study were carried out in a partially contact case (Fig. 1 **c**). Then the lower FS plate with the reagents is placed on the lower BN TC plate and the spacer is positioned on top of it, which is then covered with the upper FS plate. It is worth noting that by mixing Ge splinters or Ge powder with GeO_2 powder in the above-mentioned ratio of 1:2 and grinding them in a mortar until a macroscopically homogeneous mass is obtained, the growth of needles is also possible. But in this case, the reaction between Ge and GeO_2 proceeds so intensively that a liquid Ge- GeO_2 alloy which will be discussed below comes into contact with the upper FS plate and in situ observation of the process in the cell becomes impossible. In a cell with only Ge or only GeO_2 no needles could be grown, indicating the need for a reaction between the two components in order to grow the needles.

The assembled cell is then capped and fixed with all the remaining components shown in Fig. 1 **a**. Finally, the whole micro reactor is placed in a vacuum chamber. In it, the cell procedure for subsequent experiments takes place in two stages. First, the remaining gases are evacuated out of the cell at room temperature using low pressure of 10^{-7} – 10^{-6} mbar generated with a turbomolecular pump backed by a scroll pump over

several hours. Evacuation is possible as long as the softening temperature of the spacer is not reached, because the cell is not tightly sealed despite of polished surfaces. Subsequently, the cell is sealed in a second step by heating to a temperature of 973–1073 K. Above this temperature the soft spacer is adhered to the FS plates, causing the sealing of the cell and enclosing all products generated in reactions to be investigated afterwards.

2.3. Heating and cooling procedures

In the present study, only the lower heater was used for heating the cell, while the power supply for the upper heater was completely turned off during all the experiments. This provided a temperature of the lower FS plate which was always higher than the temperature of the upper one. Throughout the whole heating and cooling procedures, the changes in solid and liquid phases in the reaction chamber were observed *in situ* and recorded by means of a video microscope. The temperatures on the lower and upper FS plates during heating and cooling are shown in Fig. 2. In this experiment a temperature up to 1253 K on the lower FS plate was reached.

The heating and cooling procedures, presented in Fig. 2, can be divided into three stages 1–3, which will be discussed in the following text. A total of three temperature ranges (i)–(iii) were identified from analysis of a series of experiments with Ge splinters and GeO₂ powder. The processes characteristic for a certain specific range were observed in all experiments only between its minimum and maximum temperatures, which set boundaries of this range. Since the reagents lie on the lower FS plate, everywhere by default, the temperature of the lower plate is used as temperature in the following discussions. In the case of Fig. 3 illustrating the needle growth, both upper and lower FS plate temperatures are given for completeness and designated as upper TC and lower TC respectively.

Stage 1: Heating of the cell

During the first stage, only the heating of the cell takes place and no chemical reactions are observed. It takes about 7.5 min to reach 1138 K on the lower FS plate, i.e., the temperature that marks the lower limit

below which no reactions occurred in our study. The temperature difference between the lower and upper FS plates $\Delta T_{1,2}$ at the end of the first stage is approx. 120–140 K.

Stage 2: Melting of Ge splinters and of Ge-GeO₂ alloy

The second stage begins with the temperature range (i), which is between 1138 K and 1221 K as shown in Fig. 2, and is characterised by the melting of Ge splinters and of a Ge-GeO₂ alloy. The melting point of Ge equals 1211.40 K and hexagonal GeO₂ melts at 1389 K [25]. However, a eutectic point in the Ge-O phase diagram has been reported [6,7,12] according to different sources to be found at 1127 K [12] or 1143 K [6]. The existence of the eutectic explains melting point depression of GeO₂. Using the Ge-O phase diagrams presented in [6] and [12], it is possible to deduce the amount of Ge required to melt hexagonal GeO₂ by reading $x_{\text{GeO}_2(\text{hex})}$, the mole fraction GeO₂, in the temperature range 1138–1221 K. According to Dinsdale et al. $x_{\text{GeO}_2(\text{hex})} = 0.77\text{--}0.85$ [12]. Since in [6] the eutectic lies at 1143 K, $x_{\text{GeO}_2(\text{hex})} = 0.9\text{--}0.94$ in the temperature range 1143–1221 K. Although there is a huge difference in the values obtained from these two references, they both show that a sufficiently large amount of Ge is needed to melt the Ge-GeO₂ alloy at lower temperatures.

The order in which Ge splinters and Ge-GeO₂ alloy melt depends on the arrangement of the reagents, four different cases of which are shown in Fig. 1 b–e. In the full and partially contact case, the melting point of the alloy is below or at highest equals that of the pure Ge, i.e., between 1138 K and 1211.40 K [25]. When the reagents are not in contact with each other and separated by a distance (non-contact case), the alloy melts at higher temperatures of 1218–1221 K. Once one splinter is brought in contact with the powder, but other stay lying on the distance to it (quasi-non-contact case), the order of melting can be influenced: sometimes alloy melts below and sometimes above the melting point of Ge. The decisive factor on the Ge-GeO₂ alloy melting point is the distance between Ge splinters or droplets and powder. Therefore, the transfer of Ge from splinters or droplets to powder appears to take place through the gas phase. From the results we can conclude that the main reaction responsible for this is at least the partial transformation of solid (s) or liquid (l) Ge into a gaseous (g) state either through

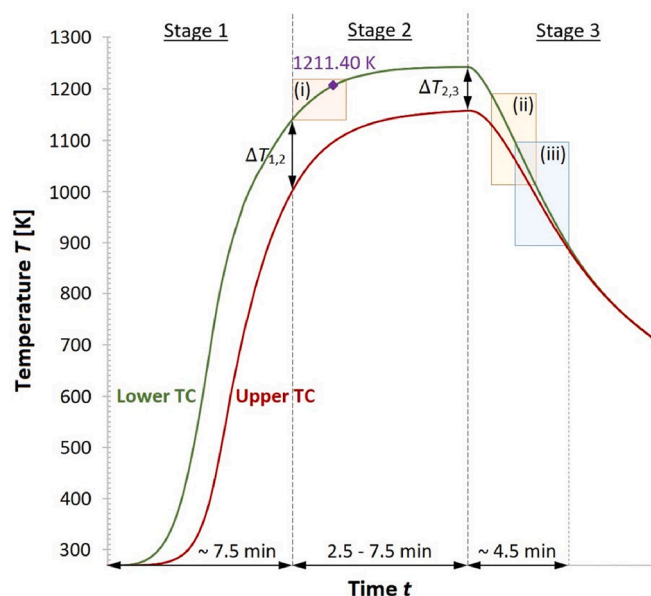


Fig. 2. The lower and upper fused silica (FS) plate temperature during cooling and heating procedure as a function of time. The temperature profile can be divided into three stages: stage 1 (heating of the cell), stage 2 (the maximum temperature is reached slowly), stage 3 (cooling of the cell). A total of three ranges can be distinguished: (i) melting of Ge splinters and of Ge-GeO₂ alloy (1138–1221 K), (ii) solidification of Ge droplets (1194–1016 K), (iii) growth of needles (1100–897 K). ΔT indicates the temperature difference between the lower and upper FS plates at the boundary between two stages: $\Delta T_{1,2}$ is approx. 120–140 K, $\Delta T_{2,3}$ is about 100–120 K after 2.5 min or 70–90 K after 7.5 min. The melting point of Ge of 1211.40 K [25] is given for reference.

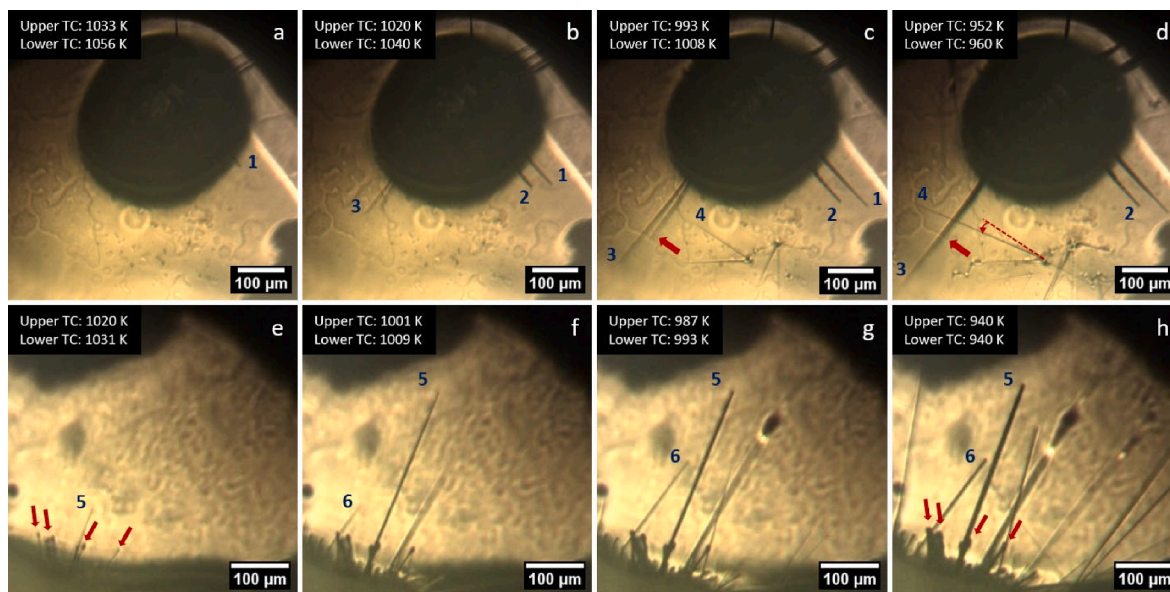


Fig. 3. **a-d** The longitudinal growth of the needles 1–4. The time after the beginning of the third stage for each image in the sequence **a-d** is 138 s, 148 s, 168 s and 198 s respectively. The red marks (arrows and dashed line) in **c** and **d** are for the purpose of explaining the tilt of needles at their growth source point (see text). **e-h** The lateral growth of the needles 5 and 6. The respective times after the onset of the third stage for images **e-h** are 157 s, 171 s, 181 s and 217 s. The red arrows in **e** and **h** indicate needles that only grow in thickness after reaching a short length and have pronounced large tips (see text). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

sublimation or through melting and subsequent evaporation



It is worth noting that neither in the full and partially contact, quasi-non-contact nor non-contact case is there any correlation between the melting point of the alloy with both the number of Ge splinters (from four to eight) and their lowest, greatest and average mass (from 25 μg to 70 μg) in a particular experiment.

The duration of the second stage is between 2.5 and 7.5 min. A longer duration time of up to 7.5 min is favourable to observe the melting of Ge splinters and the complete consumption of GeO_2 powder for the formation of a liquid Ge-GeO_2 alloy. Shorter duration of only 2.5 min allows some of the powder to be kept solid to study the possibility of needle growth on it. At the end of the second stage the temperature difference between the lower and upper FS plates $\Delta T_{2,3}$ is approx. 100–120 K after 2.5 min and respectively 70–90 K after 7.5 min.

Stage 3: Solidification of Ge droplets and growth of needles

Within the third stage the cell is cooled down from the maximum temperature of approx. 1253 K to room temperature. In the temperature range (ii), presented in Fig. 2, which is bounded by 1194 K and 1016 K, undercooled Ge droplets solidify on the lower FS plate. From 1100 K onwards needles begin to grow on the solidified droplets. Their growth stops at 897 K. In Fig. 3 **a-d** the longitudinal growth of four needles is presented. The needles 1–3 are on the large solidified droplet, which was initially a Ge splinter, and the needle 4 is on a frozen microdroplet, which was probably a small piece of a splinter. Both droplets are surrounded by a transparent thin yellowish layer of molten Ge-GeO_2 alloy. The thickness of needles is about 5–10 μm . The needles 1 and 2 are the thickest and the needle 4 is the thinnest. The needle 3 is of medium thickness. Moreover, the needles 1 and 2 grow more slowly than the two other needles, as can be seen by comparing their length increase in Fig. 3 **a-d**. The final lengths of the needles 1–4 after 59 s (the time interval between Fig. 3 **a** and **d**) are 217 μm , 144 μm , 390 μm and 402 μm respectively.

At lower temperatures, when the needle growth slows down considerably, a tilt of needles at the point from which they started to grow can be observed sometimes. Such situation can be seen for the

needles 3 and 4 in Fig. 3 **c** and **d**. The needle 3 and its neighbour, marked with thick red arrow, turn a small angle towards each other, so that in the view from above they appear overlapping one another almost completely. The needle 4 is tilted by about 10° from its original direction in Fig. 3 **c**, shown by the dashed red line in Fig. 3 **d**.

Needles can grow not only in longitudinal but also in lateral direction as shown in Fig. 3 **e-h**. For example, the needles 5 and 6, both with the starting thickness of about 3.0–3.5 μm (Fig. 3 **e** and **f**), grow laterally. The final thicknesses of the needles 5 and 6 on Fig. 3 **h** are 10 μm after 46 s from the start of growth in lateral direction and 5 μm after 50 s respectively.

Red arrows in Fig. 4 **e** and **h** mark needles which reach a length of 50–60 μm that, compared to the lengths of other needles, is much shorter, and then grow only in lateral direction. It can be observed how

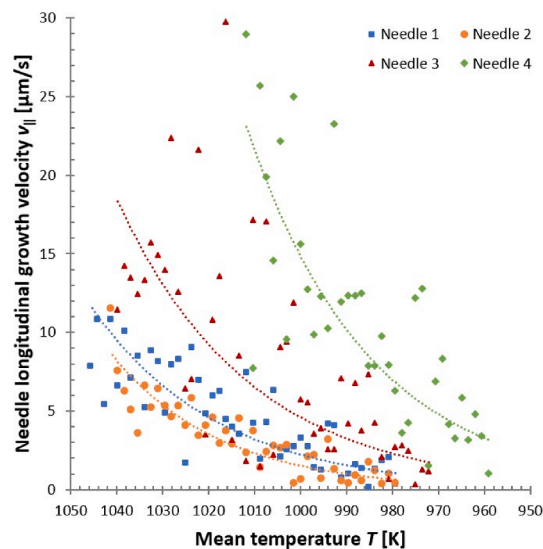


Fig. 4. The longitudinal growth velocities of needles 1–4 from Fig. 3 **a-d** as a function of mean temperature in the cell. The dashed lines mark exponential trend lines, which are plotted on the individual growth velocities.

tips of these needles are getting larger during growth.

If the duration of the second stage is reduced to 2.5 min, a small amount of the powder is still solid and needles grow on it. Since needles growing on frozen Ge droplets and as well on GeO₂ powder have the same geometry and composition, their nucleation seems not to depend much on the crystalline properties of the surface on which they are formed. Since it is impossible to predict a specific temperature point at which needles begin to grow, as well as a location and direction of needles that have grown in a particular experiment, the growth of needles on GeO₂ powder could not be directly observed. Therefore, the exact temperature range in which this process takes place could not be determined in our experiments.

3. Results

3.1. In situ microscopy of needle growth

The lengths of needles 1–4 already presented in Fig. 3 a–d were measured every second and from the results the longitudinal needle growth velocities $\nu_{\parallel}$ were calculated. Fig. 4 shows the dependency of the growth velocities on the temperature in the cell. Since the needles can grow almost parallel to the surface of the FS plate, just as all the needles in Fig. 3 did, or at an angle to it, and in some cases even reach the upper FS plate, the temperature in the cell was taken as its arithmetic mean value, calculated from the temperature of the lower and upper FS plates. For better visibility, exponential trend lines, which are marked with a dash, are plotted on the individual growth velocities. The initial growth velocity of the thin needle 4 is approx. from two to four times higher than that of the thicker needles 1 or 2, and equals 25–29 $\mu\text{m/s}$ and 7–12 $\mu\text{m/s}$, respectively. Both the thickness and the longitudinal growth velocity of the needle 3 lie between the respective values of thinner and thicker needles. Therefore, there seems to be an inverse relation between the thickness and the growth velocity of the needle.

It should be noted that if the cell is cooled according to the time–temperature diagram in Fig. 2 to a constant temperature within the range in which needle grow, i.e., between 1100 K and 897 K, the needle longitudinal growth stops once this temperature is reached and kept constant.

From measuring the thickness of the needles 5 and 6 presented in Fig. 3 e–h, their lateral growth velocity $\nu_{\perp}$ can be calculated. The initial $\nu_{\perp}$ of the needle 5 is approx. 0.11 $\mu\text{m/s}$ at 1000 K and that of the needle 6 is about 0.05 $\mu\text{m/s}$ at 990 K. Therefore, at a temperature of 990–1000 K $\nu_{\perp}$ is by one to two orders of magnitude smaller than $\nu_{\parallel}$. The lateral growth velocity also depends on the temperature and decreases during cooling.

In general, two limiting cases of needle growth can be considered. According to the first one, needles grow in both directions, longitudinal and lateral, simultaneously, as shown in Fig. 3 a–d. It can be seen that the thickness of the base of such needles, e.g., of the needles 1 and 2, is bigger than that of their tip. Measurements of the tip thicknesses of the needles 1 and 2 at the beginning and at the end of the longitudinal growth process, as presented in Fig. 3 a, b and d respectively, show no noticeable change in their thickness. Depending on the longitudinal growth velocity, the difference between thickness of needle base and tip can be visible clearly, as in the case of needles with lower values of $\nu_{\parallel}$ (needle 1 and 2), or less noticeable, as it is for needles with higher $\nu_{\parallel}$, which grow very rapidly in length (needle 4). According to the second mechanism, needles reach a certain length, stop growing in longitudinal direction and increase only laterally. Their thickness is about the same along their entire length, as shown in Fig. 3 e–h. So, the needles can reach small lengths and have pronounced large tips (needles marked with red arrows), or they can be as long as needles, growing according to the first mechanism, and not have any noticeable thickening at the tips (needles 5 and 6).

3.2. Ex situ analysis of needle surface structure and cross-section

When 573 K is reached, the vacuum system is turned off. Once ambient pressure and temperature inside the vacuum chamber is established, the cell can be removed from it. To examine the sample the cell must be opened. Therefore, the circular notch which was grinded in one of the FS plates, as shown in Fig. 1, serves as a crack line when mechanical stress is applied. It is simply possible to open the cell by slightly breaking a small hole in the notch with the tip of tweezers, then pick up the small platform-like area with the sample and break it out. Subsequently, the sample is placed on a holder, coated with carbon, and analysed with FE-SEM (JEOL JSM-6700F) as well as EDX spectroscopy combined with element mapping (Oxford Instruments INCA 300).

According to SEM analysis, most of the needles on frozen Ge droplets are between 200 μm and 400 μm in length. In some cases, shorter needles up to 100 μm or longer ones being almost 450 μm were also found. The needles on the GeO₂ powder are up to 100 μm in length. As can be seen in Fig. 5 a–f, the needles do not exceed about 10 μm in thickness; the thinnest, such as in Fig. 5 e and f, are approx. 0.25 μm thick. Also, the needles can have different surface structure, as shown in Fig. 5 a–f, but they all exhibit similar polyhedral shapes, which form a kind of base unit for the structure surface. Their edges, highlighted by a thick line, show that these polyhedral structural microunits can be densely or quite loosely packed along the needle length. In the latter case their three-dimensional shape is easily discernible, e.g., in Fig. 5 d. There are three main types of highly regular arrangements of the microunits: either in a single row as in Fig. 5 a and f, or in two parallel rows with or without overlap, as in Fig. 5 b and c respectively. The structure of some needles exhibits a disturbance of the strict ordering: in Fig. 5 d there is a non-uniformity in the size of the polyhedra arranged in a row, while in Fig. 5 e some polyhedra are not only increased in size compared to others, but there is also a non-uniform overlap of them.

The tips of the needles are also varied, as shown in Fig. 5 g–k. It should be, however, directly mentioned that these tips do not represent small balls often observed in vapor-liquid-solid (VLS) or catalytic tips typical for vapor-solid-solid (VSS) growth mechanisms. The growth of the tips can be observed in situ, e.g., in Fig. 3 e–h they are marked by red arrows, but their role in needle growth is unclear. In Fig. 5 g, the tips are the source of growth for new needles, or are bifurcated. They can also appear in quite messy ways, such as in Fig. 5 h, or they can show some regularities, e.g., edges highlighted in Fig. 5 i, similar to the polyhedral microunits on the surface of needles. In some cases, the tips are very pronounced and geometrically regular and have a shape of a truncated triangular trapezohedron (Dürer's solid) or of an octahedron, as shown in Fig. 5 j and k respectively. These both shapes are easily discernible in the surface structure of all needles presented in Fig. 5.

The needles are quite fragile and some of them, especially the long ones, break easily after opening the cell. Broken needles expose their cross-sections, which is a valuable material for studying their internal structure and composition. Fig. 5 l–o shows cross-sections of such needles. Clearly distinguishable is the core, which looks like an irregular hexagon, with three sides longer than the other three sides, as is highlighted in Fig. 5 n. The needle shell is made up of six branches that originate from each vertex of the core and the darker mass surrounding them. This can be readily seen in Fig. 5 l and m. Moreover, there is noticeable variation in the shell-to-core area ratio in the different needle cross-sections. So, in the cross-sections of the needles in Fig. 5 l and m, the area occupied by the shell is much larger than that of the core, which implies a higher value of the shell-to-core area ratio. But this ratio has smaller values in the case of the needles in Fig. 5 n and o.

Fig. 6 shows the results of EDX spectroscopy combined with element mapping study of the needles. A close look at the surface structure in Fig. 6 a and b clearly shows the contrast between the faces (position 1 and 3) and the lighter edges (positions 2 and 4) of polyhedral structural microunits. The cross-section of the broken needle also shows structures of different contrasts: the core (position 6) with its six branches and their

surrounding material (position 5). Fig. 6 c contains low-energy part of two EDX spectra taken at positions 1 and 2 with acceleration voltage of 10 kV, which are normalised with respect to the Ge-L α peak. In addition to Ge and O, from which the reagents – Ge splinters and GeO₂ powder – are composed, C and Si are also present on the spectra. C is due to the carbon coating of the samples. Si probably come from the Corning® glass spacer, which turns soft at high temperatures, or can be understood

in terms of weak reactions of Ge and GeO₂ with the FS plates. However, no other elemental constituents of the spacer such as Al, B, Ca, Mg and Sr were detected. From the comparison of the spectra, it can be seen that the edge consists mainly of Ge, while the face composition contains an even higher amount of O. The same results were obtained when analysing other faces and edges (e.g., positions 3 and 4) of the surface structures as well as when examining the core of the needle and the

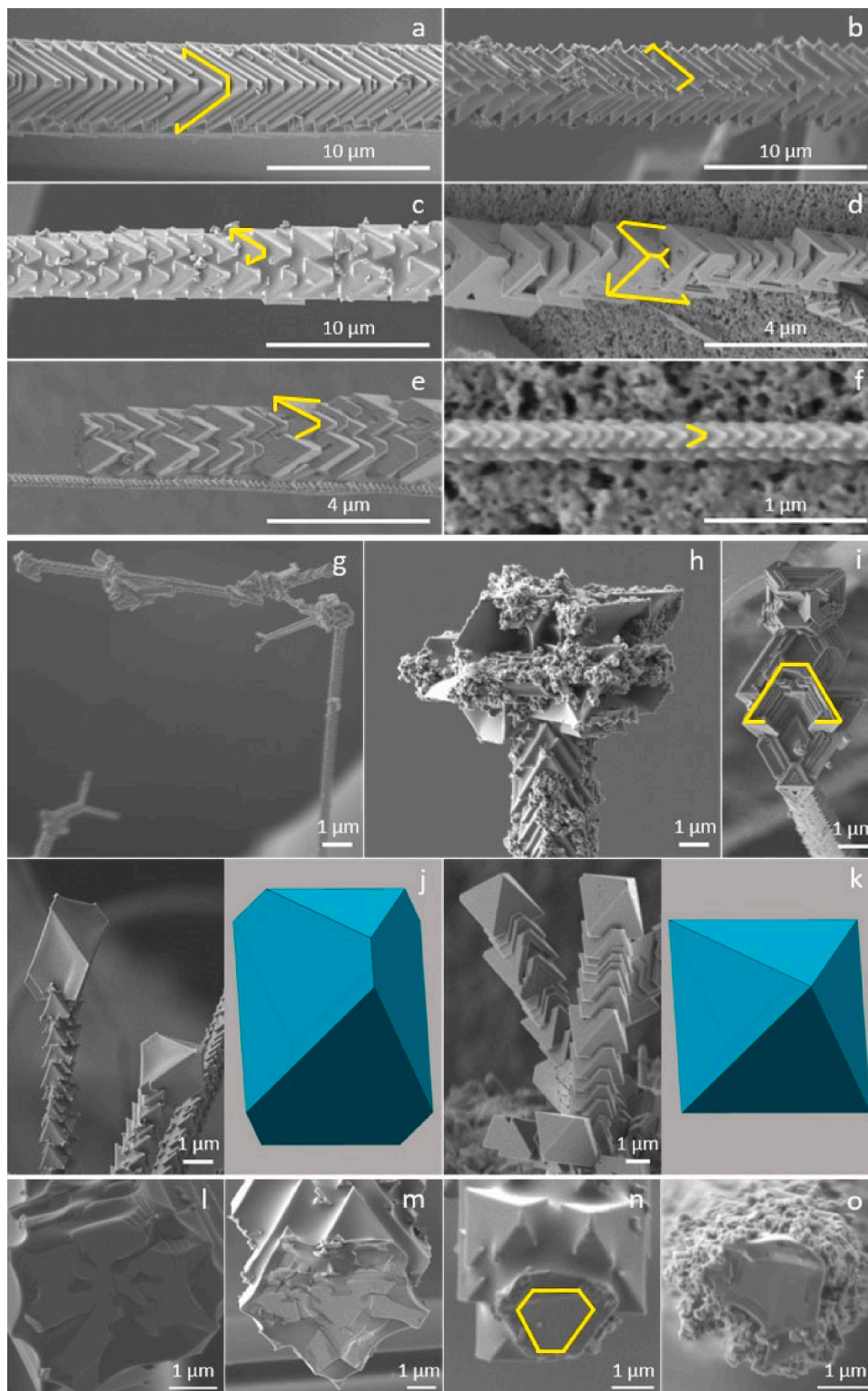


Fig. 5. a-f Different surface structures and g-k tip forms of needles, and l-o cross-sections of broken needles as explained in the text.

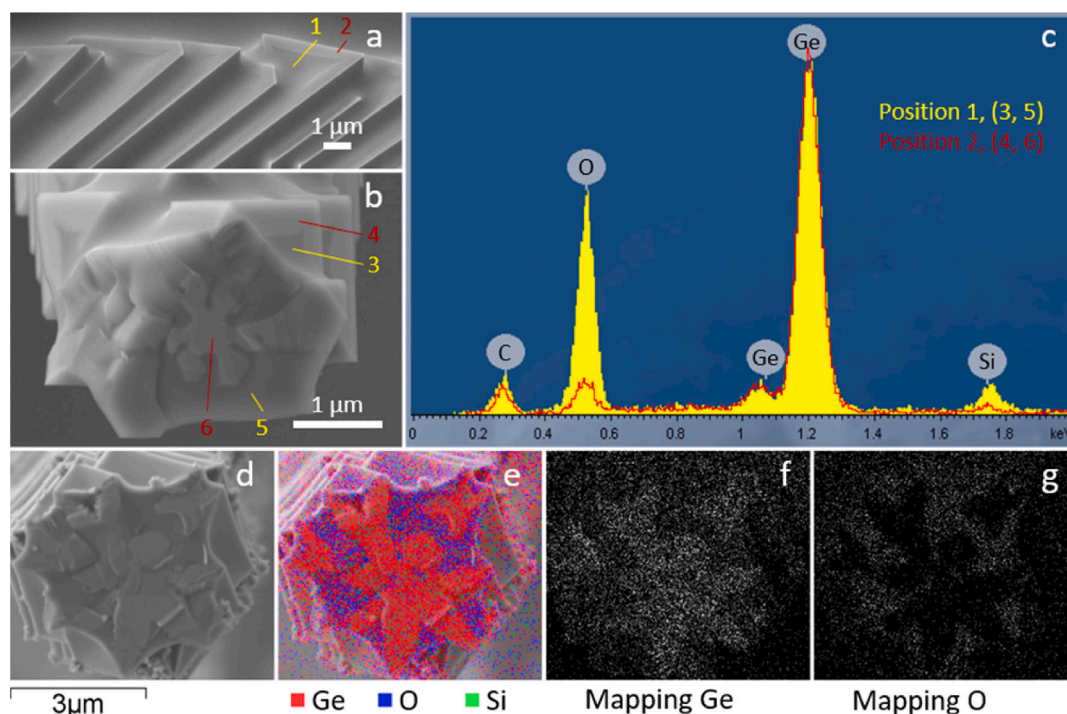


Fig. 6. **a** The surface and **b** the cross-section of needle. **c** The EDX spectra taken at positions 1 and 2. Almost the same spectra were obtained when analysing positions 3–6. **d** The cross-section of needle at which the EDX mapping was done. The results of the mapping **e** with all elements, from which the needle is composed, **f** with only Ge intensities and **g** with only O intensities.

surrounding material (positions 6 and 5 respectively). Therefore, the core branches extend to the surface of the needle, where they form the edges of the polyhedral structural microunits (position 2 and 4). Also, the mass, which surrounds the core with its branches, extends to the surface and forms there the faces of the polyhedra (positions 1 and 3).

The result of the needle cross-section EDX element mapping with acceleration voltage of 5 kV is presented in Fig. 6 d–g. It confirms that the core and branches consist mainly of Ge, as is evident from the strong Ge intensities in Fig. 6 f and weak O intensities in Fig. 6 g from these areas. The same can be seen in the mapping with all elements in Fig. 6 e, which also shows that Si is present in both the core with its branches and the mass surrounding them. It can, therefore, be concluded from EDX spectroscopy combined with element mapping that the edges and the core with its branches are composed mainly of Ge (Ge-rich phase), while the remaining filling mass of the needle is made of Ge and O (Ge- and O-rich phase). To quantify the ratio of atomic percentages of O to Ge ($\text{at}\%[\text{O}]:\text{at}\%[\text{Ge}]$) in the latter phase, the EDX measurements were calibrated with a hexagonal GeO_2 powder and FS plates. It remains unclear whether Si is incorporated into the Ge- and O-rich phase either as elemental Si or as a chemical compound, e.g., SiO_2 . Two estimates can therefore be made: with Si and with SiO_2 as impurity. For the faces on the needle surface, i.e., for positions 1 and 3 in Fig. 6 a and b respectively, $\text{at}\%[\text{O}]:\text{at}\%[\text{Ge}]$ is 2.34 in the first case (Si as impurity) and 2.01 in the second (SiO_2 as impurity).

3.3. Ex situ analysis of needle internal structure

A FE-SEM (FE-SEM-FIB Zeiss AURIGA) equipped with a gallium focused ion beam (Ga-FIB) was employed to cut longitudinally one of the needles with a polyhedron tip to study its internal structure. Fig. 7 a, b and c show the needle before, during and after preparation, respectively. As can be seen from these SEM images, the high regularity of the

arrangement of polyhedral microunits is remarkable not only on the surface of the needle, but also inside it all way down to the core. This regularity is exemplified by the two domains of the same phase (Ge-rich phase), which are highlighted in red in Fig. 7 b and c. On the latter, the needle core and the growth of branches from it are easily discernible. The dashed lines in Fig. 7 b and c mark the edges of two microunits before cutting the needle, which are indicated by the solid line in Fig. 7 a.

After preparation of the needle, its upper part with the tip, was examined by EDX mapping and spectroscopy (Oxford X-Max) with acceleration voltage of 5 kV. The results of the first of these analyses are shown in Fig. 7 d. On it, as on the cross-section of the needle in Fig. 6 e, two phases of different composition can be distinguished: Ge-rich and Ge- and O-rich phase. Si was also found in both of them: approx. 0.1–0.6 atomic percentage of Si ($\text{at}\%[\text{Si}]$) in the core with its branches and one order of magnitude more in the surrounding mass, i.e., up to 6 $\text{at}\%[\text{Si}]$. Moreover, the distribution of Si within each phase is fairly homogeneous. Apart from Ge, O and Si, no other elements, could be detected in the longitudinal section. In the same way as described above for the needle surface, the ratio $\text{at}\%[\text{O}]:\text{at}\%[\text{Ge}]$ for the Ge- and O-rich phase inside the needle was estimated. With Si and SiO_2 as impurity, it equals 2.21 and 1.79 respectively.

To study the internal structure of the needle with STEM and its crystal structure with SAED, both with FE-TEM (JEOL JEM-2100F-UHR), a segment of one needle was fixed with platinum (Pt) on a copper grid and then a thin lamella was prepared of it with a Ga-FIB. Fig. 8 a and b shows the STEM images of the lamella with a thickness of about 125 nm at the top and approx. 10 nm at the bottom. Due to the preparation process the bottom edge looks uneven. For example, there is a small groove, marked with a green arrow in Fig. 8 a and b, around which the lamella is sufficiently transparent to electrons. Yellow and red crosses with numbers from 1 to 8 mark areas in which SAED was made.

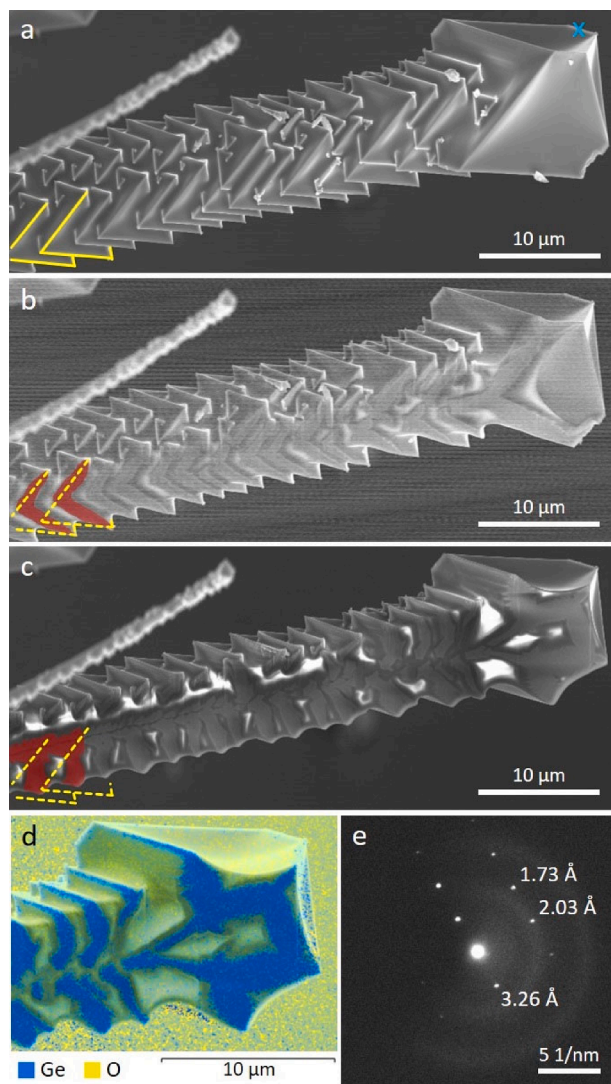


Fig. 7. Needle with polyhedral tip **a** before, **b** during and **c** after preparation of the longitudinal section. Two domains of the Ge-rich phase are highlighted in red in **b** and **c**. The dashed lines in **b** and **c** mark the edges of two microunits before cutting the needle, which are indicated by the solid line in **a**. **d** The mapping results of the needle upper part with polyhedral tip. **e** The SAED pattern obtained from the thin tip edge marked with a blue cross in **a**. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Areas 1 and 2 lie in the Ge- and O-rich phase and their diffraction patterns show no reflexes, i.e. correspond to the diffraction pattern of an amorphous solid. In contrast, areas 3 to 8 are in the core and branches, i.e. in the phase composed predominantly of Ge, and show reflexes which are typical for a crystal, Fig. 8 c. The diffraction patterns in all these areas are identical to each other, which presumably indicates that the branches grow homoepitactically on the core. Many of the branches to the right of area 8, as can be seen in Fig. 8 a and b, grow directly from the core. This can also be seen from the needle cross-sections shown in Fig. 5 l-o and Fig. 6 b and d. They also exhibit that the branches originate from the small area on the core vertices and widen as they grow towards the surface. Therefore, the small gaps between some branches, e.g., those marked with 5, 6 and 7 in Fig. 8 b, and the core could be probably explained by the deflection of the FIB slicing line by a small angle relative to the needle core.

Another SAED pattern, shown in Fig. 7 e, was also obtained from the thin tip edge of the needle. Such an area is marked with a blue cross in Fig. 7 a. From the analysis of this diffraction pattern three spacings between adjacent lattice planes follow: 3.26 Å, 2.03 Å and 1.73 Å. According to PDF 01-073-7012, which was calculated from inorganic crystal structure database (ICSD) using POWD-12++ and data from [27], these spacings respectively correspond to the planes (111), (220) and (311) of a cubic Ge, space group *Fd-3m*, No. 227. The same result is obtained from the diffraction pattern in Fig. 8 c, from which the following two atomic spacings were calculated: 3.26 Å and 1.71 Å.

The analysis of the lamella through EDX spectroscopy in the FE-TEM (Oxford Instruments INCA 200) with acceleration voltage of 20 kV confirmed previous results about the homogeneous distribution of Si and its atomic percentages in both phases inside the needle. Similarly, only Ge, O and Si are the elements that could be detected in it. With Si and SiO₂ as impurity in the Ge- and O-rich phase, at%[O]:at%[Ge] equals 2.25 and 2.00 respectively.

Needles with highly regular arrangements of the microunits like the one shown, e.g., in Fig. 7, were studied by Raman spectroscopy (Olympus BX microscope with Raman spectrometer Bruker Senterra). Due to the laser spot size being comparable to the thickness of the needles themselves, considering its penetration depth into material in μm-range and the structural and compositional heterogeneity of the needles, Raman spectroscopy could not be performed only on the Ge-rich or on the Ge- and O-rich phases. Therefore, the information obtained by spectroscopy was gained from a sufficiently large volume of material. The result is a single pronounced peak at approx. 297–299 cm⁻¹. According to [28], the theoretical and experimental values of 299 cm⁻¹ and 301 cm⁻¹ respectively correspond to the F_{2g} optical mode of a cubic Ge. This is therefore in complete agreement with the SAED results.

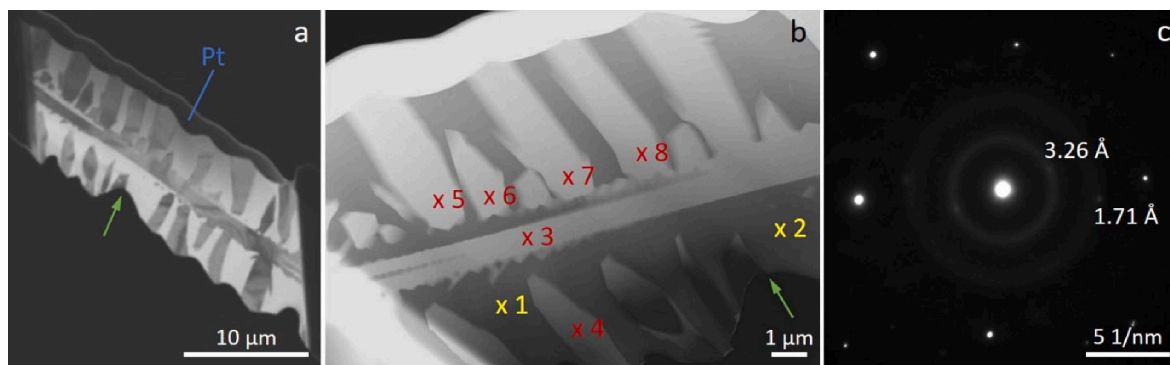
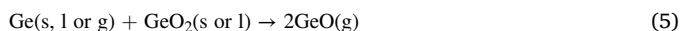


Fig. 8. **a, b** STEM image of a prepared lamella fixed with platinum (Pt) on a copper grid (not shown). Numbers 1 to 8 mark areas, located around a small groove marked with a green arrow, in which the SAED was made. **c** SAED pattern of areas 3 to 8. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

4. Discussion

It is difficult to propose a mechanism for the growth of these complex needle structures. A formation via the gas phase might involve Ge(g) as well as GeO(g). During the second stage of heating, significant quantities of GeO gas are produced through



Our thermodynamic calculations using the Thermodynamik (Max Planck Institute, CPfS) software and data from [26] show that the GeO gas has the highest partial pressure among other gases in the cell. Its value at temperatures in the range bounded by 1138 K and 1253 K, i.e., the temperature at which the second stage begins and the highest temperature reached in our experiments respectively, is approx. 27–240 mbar. However, there is also a partial pressure of $(0.12\text{--}4.25) \times 10^{-6}$ mbar for Ge gas. When the cell is cooling down during the third stage, molten Ge droplets solidify and needles begin to grow on them, as shown in Fig. 2. The needle growth takes place in the temperature range (iii), i.e. between 1100 K and 897 K. As our thermodynamic calculations show, the partial pressure of GeO gas at 1100 K is approx. 12 mbar, and at 897 K it is only about 0.05 mbar. Within the range (iii), the temperature in the cell lowers by about 200 K, which occurs in 2–2.5 min, and within this time, the partial pressure of GeO drops by two orders of magnitude. A decrease in temperature can lead to intensive deposition of GeO, which then might disproportionate into solid Ge and GeO₂ [4,5,17,22]. Therefore, reaction (2) followed by disproportionation (3) could be involved in needle formation and growth. Assuming complete disproportionation of solid GeO would explain needles composed of Ge and GeO₂. This is consistent to our SAED and EDX spectroscopy results of the needle surface and internal structure. The growth mechanism itself remains unclear.

5. Conclusions

A method for growing composite germanium/germanium oxide needles with thicknesses of 0.25–10 µm and lengths up to 450 µm in the range of 897–1100 K was presented. Moreover, their growth was observed in situ by means of a video microscopy. The longitudinal needle growth velocity has been determined and its dependency on temperature and needle thickness has been shown. Surface and internal structure of the needles were analysed ex situ with SEM, EDX spectroscopy combined with element mapping, STEM, SAED and Raman spectroscopy, and the results were discussed.

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CRediT authorship contribution statement

Alexey S.T. Rybakov: Conceptualization, Methodology, Validation, Formal analysis, Investigation, Writing – original draft, Visualization. **Lea L. Meyer:** Validation, Investigation, Writing – review & editing. **Henning Kuper:** Conceptualization, Investigation, Writing – review & editing. **Mark Chalupa:** Investigation. **Verena Becker:** Conceptualization, Resources, Writing – review & editing. **Jörg August Becker:** Conceptualization, Methodology, Resources, Writing – review & editing, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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